

Novelty Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Novelty, Transience, and Resonance Analysis

PENPAL Project

2026-05-06

Contents

1	Overview	1
2	Load Data	2
3	Preprocessing	2
4	Primary: Within-Condition Slot Slope-Interaction Asymmetry	2
5	Bootstrap CIs and Effect Sizes on A_influence_story	4
6	Planned Contrast: HA vs (HH + AA)/2	4
7	Directional Stability of Signed Novelty-Resonance Slope Asymmetry	4
8	Length-Controlled Variant of A_influence	5
9	Supplemental: General Condition-Level Comparison	5
10	Supplemental: Pooled Novelty-to-Resonance Relationship	8
11	Supplemental: Role-Level View	10
12	Summary	10

1 Overview

This notebook compares three information-theoretic metrics across conditions:

- **Novelty / surprise**: how unexpected a contribution is given prior context
- **Transience**: how quickly its influence fades
- **Resonance**: novelty minus transience

The notebook now follows the new data structure:

- it loads novelty scores through the shared comparison helpers,
- it restricts the analysis to **complete exchanges** in the aligned turn window,
- it treats `author_1` and `author_2` as **slot 1 / slot 2** rather than fixed identities,
- it uses role labels only where they are structurally valid:
 - **Human / AI** in Human-AI
 - **Starter-side / Non-starter-side** in same-type conditions

- role-labeled plots are therefore **within-condition descriptive views**, not direct cross-condition comparisons

Primary cross-condition object. Following the paper’s narrative-influence model, the main inferential result is the **within-condition novelty-to-resonance slope interaction** between the two slots, fit as $\text{resonance} \sim \text{surprise} \times \text{slot} + (1 \mid \text{conversation_id})$ per condition. The cross-condition asymmetry magnitude is

$$A_{\text{influence}}(\text{condition}) = |\beta_{\text{surprise, slot=1}} - \beta_{\text{surprise, slot=2}}|,$$

extracted via **emtrends**. Per-story OLS slope differences provide a label-invariant robustness layer. Bootstrap CIs, Cliff’s δ effect sizes, the planned HA vs (HH+AA)/2 contrast, and a length-controlled re-fit are added below. The signed Human/AI decomposition lives in analysis/human-ai/novelty.Rmd and is not duplicated here. Starter-mechanism analysis is deferred.

Article-use note. The article-facing result is the magnitude of the novelty-to-resonance slope gap. Do not infer Human-AI specificity from within-condition significance alone; the direct cross-condition question requires comparing the slot-slope gap across conditions and checking the per-story absolute gap as a robustness layer.

2 Load Data

Table 1: Loaded novelty data by condition

condition	n_stories	n_turns
ai-ai	80	640
human-ai	96	667
human-human	36	288

3 Preprocessing

Slot-level rows: 3034

Metric rows: 9102

4 Primary: Within-Condition Slot Slope-Interaction Asymmetry

Table 2: Primary: $|\text{slope_1} - \text{slope_2}|$ from $\text{resonance} \sim \text{surprise} *$ slot per condition

condition	slope_slot1	slope_slot2	signed_diff	se	t	p	A_influence	sig
Human-AI	0.989	0.945	-0.044	0.051	-0.867	0.386	0.044	
Human-Human	0.936	0.994	0.058	0.069	0.836	0.404	0.058	
AI-AI	0.906	0.941	0.035	0.033	1.047	0.295	0.035	

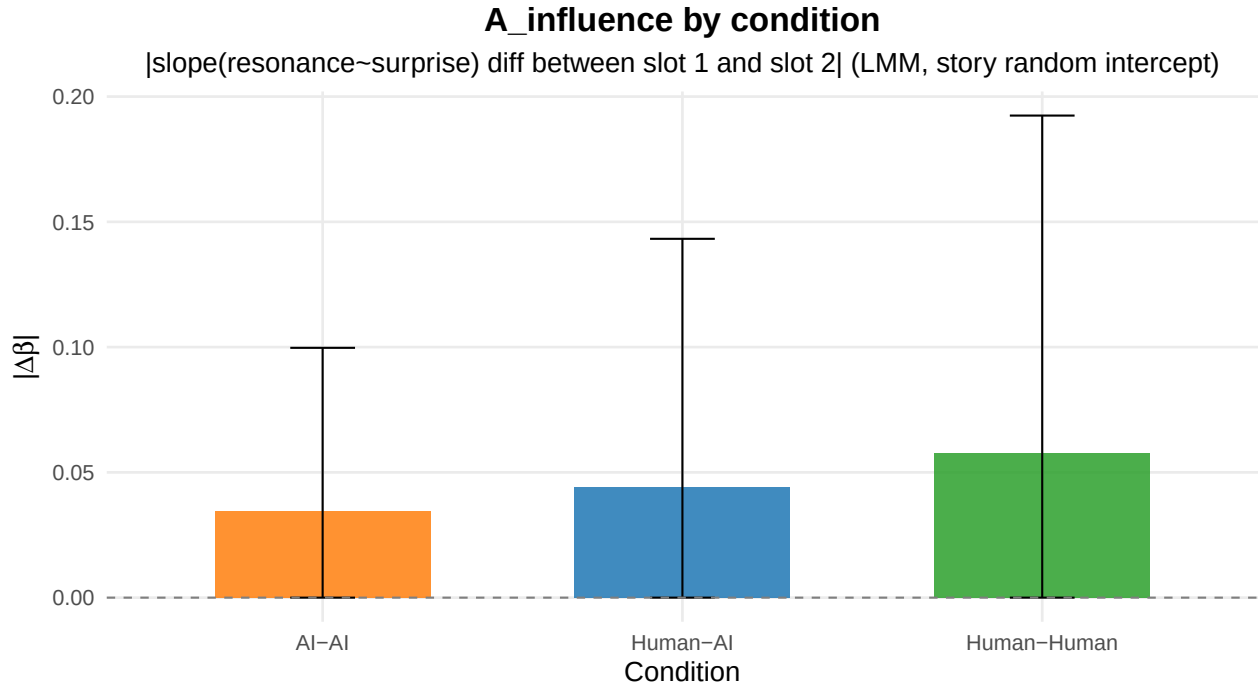


Table 3: Pooled model: |slope_slot1 - slope_slot2| (unsigned) by condition

condition	A_influence	se
Human-AI	0.066	0.045
Human-Human	0.060	0.065
AI-AI	0.041	0.042

Table 4: Robustness: per-story |slope_1 - slope_2| (OLS per story, 4 points per slot)

condition	n_stories	median_A	mean_A
Human-AI	96	0.468	0.619
Human-Human	36	0.431	0.484
AI-AI	80	0.343	0.429

```
##
## Kruskal-Wallis on per-story A_influence:
##
## Kruskal-Wallis rank sum test
##
## data: A_influence_story by condition
## Kruskal-Wallis chi-squared = 4.214, df = 2, p-value = 0.1216
##
##
## Pairwise Wilcoxon on per-story A_influence (BH-adjusted):
##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
```

```
## data: per_story_slope_diff$A_influence_story and per_story_slope_diff$condition
##
##           Human-AI Human-Human
## Human-Human 0.51      -
## AI-AI       0.15      0.34
##
## P value adjustment method: BH
```

5 Bootstrap CIs and Effect Sizes on A_influence_story

Table 5: Bootstrap 95% CI on per-condition mean of per-story A_influence (R = 10,000)

condition	n	mean	ci_lower	ci_upper
Human-AI	96	0.619	0.519	0.726
Human-Human	36	0.484	0.385	0.593
AI-AI	80	0.429	0.360	0.503

Table 6: Pairwise Cliff’s delta on per-story A_influence

contrast	n_a	n_b	cliff_delta	magnitude
Human-AI vs Human-Human	96	36	0.075	negligible
Human-AI vs AI-AI	96	80	0.171	small
Human-Human vs AI-AI	36	80	0.140	negligible

6 Planned Contrast: HA vs (HH + AA)/2

Table 7: Planned contrast: per-story A_influence in HA vs the average of HH and AA

contrast	estimate	se	t	df	p	ci_lower	ci_upper
Human-AI vs (Human-Human + AI-AI)/2	0.162	0.063	2.594	150.564	0.01	0.039	0.286

7 Directional Stability of Signed Novelty-Resonance Slope Asymmetry

Tests whether the *direction* of the slot-1 vs slot-2 novelty-to-resonance slope difference is consistent across stories within each condition. The signed quantity is $\text{signed}_i = \beta_i^{\text{slot1}} - \beta_i^{\text{slot2}}$. HA-specific identity-tied directionality predicts high stability in HA and low stability in HH/AA, even when unsigned A_influence is similar.

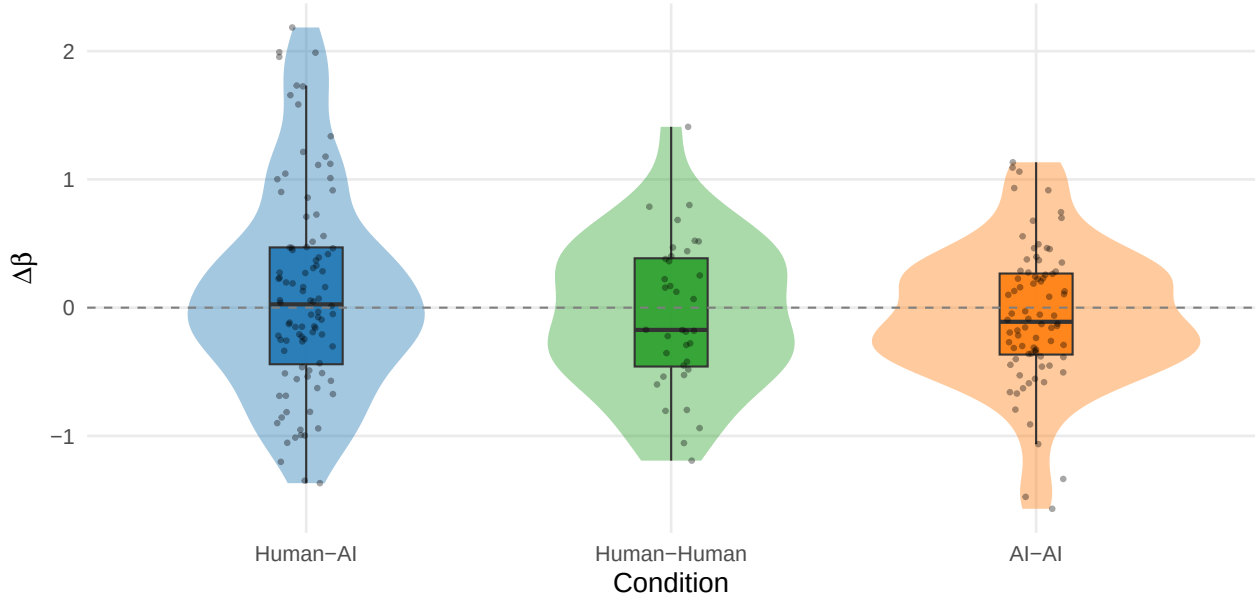
Table 8: Per-condition directional stability of signed novelty-resonance asymmetry

condition	n	mean	sd	stability	sign_consistency	one_sample_t_p
Human-AI	96	0.122	0.804	0.151	0.510	0.141
Human-Human	36	-0.053	0.583	0.091	0.528	0.589
AI-AI	80	-0.065	0.543	0.120	0.575	0.287

Brown-Forsythe variance-equality p = 0.0089

Signed novelty-resonance slope asymmetry by condition

slope_slot1 - slope_slot2 per story



8 Length-Controlled Variant of A_influence

Re-fit the within-condition LMM with turn length as an additive covariate and recompute the unsigned slope-slot magnitude. Tests whether condition-level slot-asymmetry differences survive turn-length adjustment.

Table 9: Length-controlled $|\text{slope_slot1} - \text{slope_slot2}|$ (resonance ~ surprise * slot + nwords)

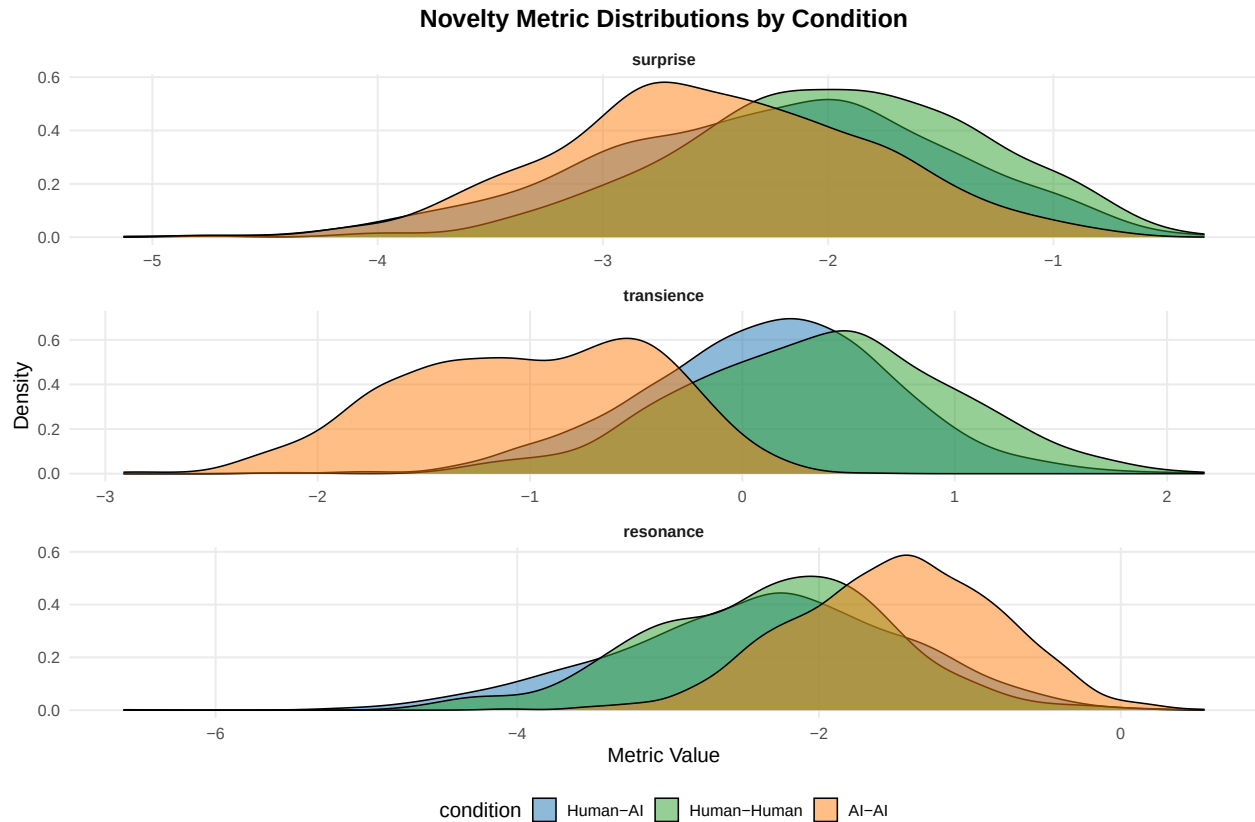
condition	A_influence_lc	se	p
Human-Human	0.057	0.069	0.403

9 Supplemental: General Condition-Level Comparison

Table 10: Overall novelty metrics by condition

condition	metric	n_stories	n_obs	mean_value	sd_value
Human-AI	surprise	96	1238	-2.263	0.772

condition	metric	n_stories	n_obs	mean_value	sd_value
Human-AI	transience	96	1238	0.120	0.589
Human-AI	resonance	96	1238	-2.382	0.940
Human-Human	surprise	36	557	-1.990	0.672
Human-Human	transience	36	557	0.351	0.627
Human-Human	resonance	36	557	-2.341	0.801
AI-AI	surprise	80	1239	-2.507	0.690
AI-AI	transience	80	1239	-1.004	0.594
AI-AI	resonance	80	1239	-1.503	0.679



```
##
## === surprise ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 6273.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.0555 -0.6215  0.0118  0.6619  3.0753
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## conversation_id (Intercept) 0.1051   0.3243
```

```

## Residual                      0.4146    0.6439
## Number of obs: 3034, groups:  conversation_id, 212
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    -2.26409    0.03782 221.29091 -59.857 < 2e-16 ***
## conditionHuman-Human    0.26790    0.07139 208.90454   3.753 0.000227 ***
## conditionAI-AI        -0.25157    0.05550 211.92507  -4.533 9.72e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndH-H
## cndtnHmn-Hm -0.530
## condtnAI-AI -0.682  0.361

## contrast      estimate      SE df z.ratio p.value
## (Human-AI) - (Human-Human)   -0.268 0.0714 Inf  -3.753  0.0005
## (Human-AI) - (AI-AI)         0.252 0.0555 Inf   4.533  <.0001
## (Human-Human) - (AI-AI)      0.519 0.0729 Inf   7.125  <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === transience ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 4681.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.5028 -0.5988  0.0145  0.6231  3.4359
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## conversation_id (Intercept) 0.1191  0.3451
## Residual              0.2357  0.4855
## Number of obs: 3034, groups:  conversation_id, 212
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    0.12084    0.03783 216.70273   3.194 0.00161 **
## conditionHuman-Human    0.22003    0.07185 209.75053   3.062 0.00248 **
## conditionAI-AI        -1.13734    0.05577 211.45331 -20.394 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndH-H
## cndtnHmn-Hm -0.527
## condtnAI-AI -0.678  0.357

```

```

## contrast estimate SE df z.ratio p.value
## (Human-AI) - (Human-Human) -0.22 0.0719 Inf -3.062 0.0066
## (Human-AI) - (AI-AI) 1.14 0.0558 Inf 20.394 <.0001
## (Human-Human) - (AI-AI) 1.36 0.0736 Inf 18.454 <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === resonance ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 7361.2
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -4.9528 -0.6360 0.0540 0.6672 3.4006
##
## Random effects:
## Groups Name Variance Std.Dev.
## conversation_id (Intercept) 0.03175 0.1782
## Residual 0.63563 0.7973
## Number of obs: 3034, groups: conversation_id, 212
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) -2.38343 0.02906 232.78641 -82.02 <2e-16 ***
## conditionHuman-Human 0.04391 0.05355 199.85325 0.82 0.413
## conditionAI-AI 0.88197 0.04189 207.75074 21.05 <2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) cndH-H
## cndtnHmn-Hm -0.543
## condtnAI-AI -0.694 0.376
##
## contrast estimate SE df z.ratio p.value
## (Human-AI) - (Human-Human) -0.0439 0.0536 Inf -0.820 1.0000
## (Human-AI) - (AI-AI) -0.8820 0.0419 Inf -21.055 <.0001
## (Human-Human) - (AI-AI) -0.8381 0.0542 Inf -15.473 <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests

```

10 Supplemental: Pooled Novelty-to-Resonance Relationship

```

## Novelty-to-resonance model:
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: resonance ~ surprise * condition + (1 | conversation_id)

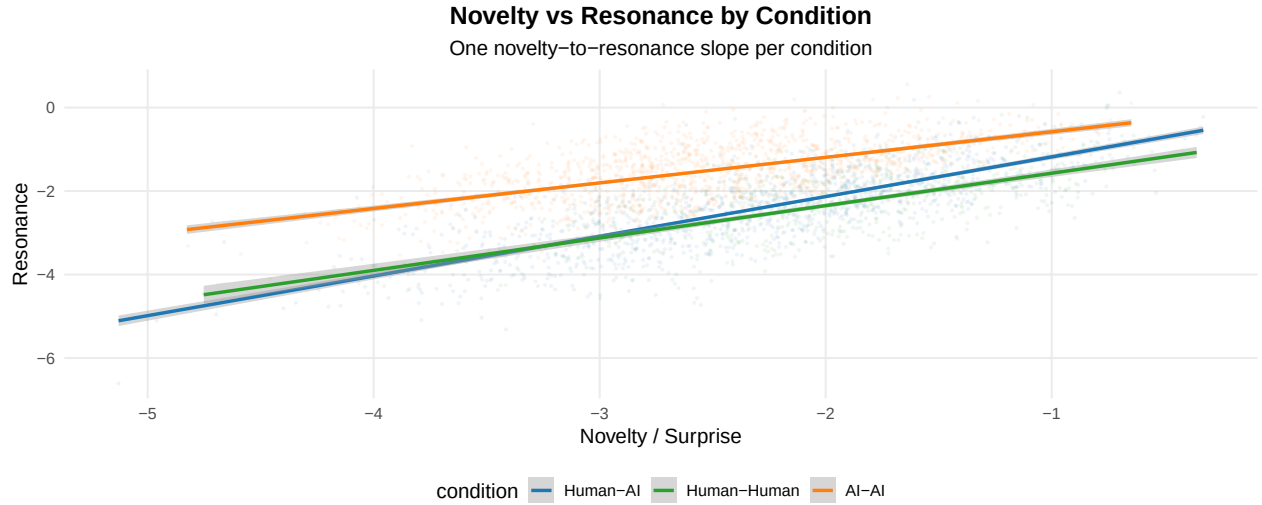
```



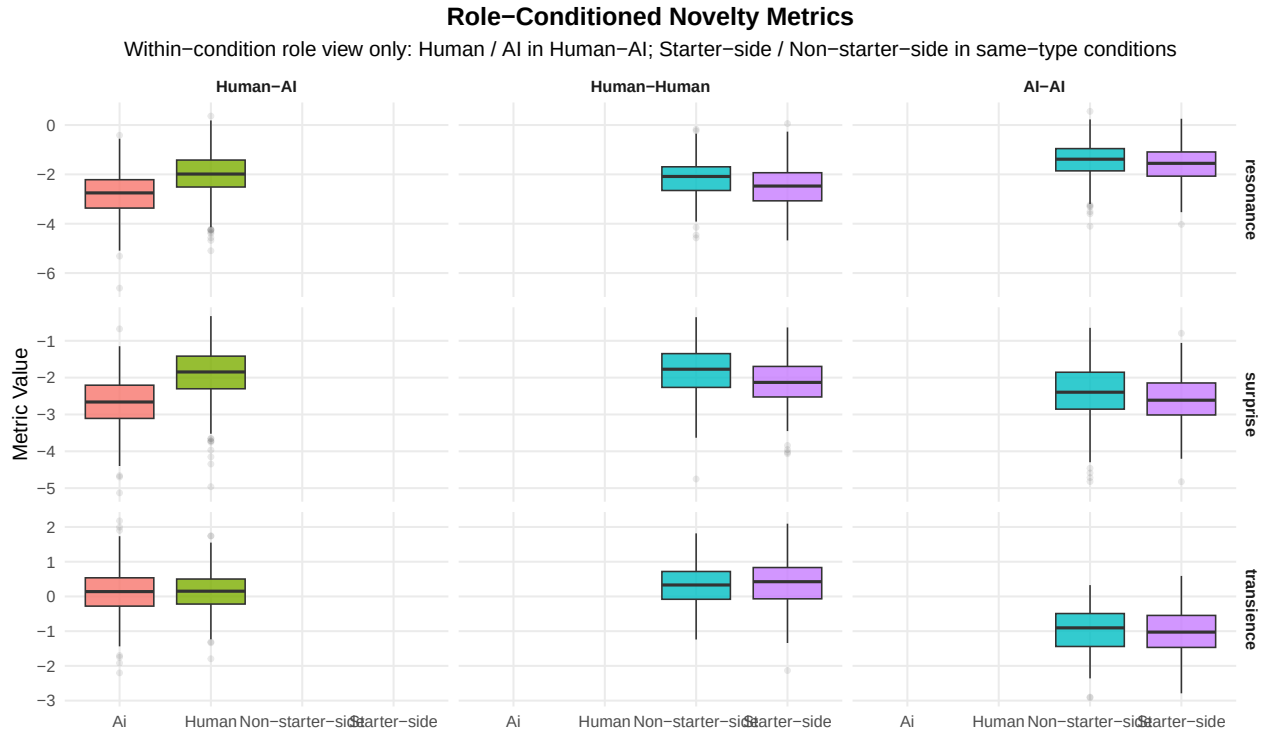
```

## Data: df_slots
##
## REML criterion at convergence: 4676.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.4456 -0.6320 -0.0074  0.6004  4.4945
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 0.1033     0.3214
## Residual                   0.2363     0.4861
## Number of obs: 3034, groups: conversation_id, 212
##
## Fixed effects:
##                                     Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)                      -0.15276    0.05598 1019.51202  -2.729  0.00647
## surprise                        0.98589    0.01908 2929.55544  51.664 < 2e-16
## conditionHuman-Human            -0.26262    0.10670  964.38386  -2.461  0.01402
## conditionAI-AI                  0.87795    0.09212 1229.37515   9.530 < 2e-16
## surprise:conditionHuman-Human   -0.02312    0.04010 2993.36197  -0.577  0.56423
## surprise:conditionAI-AI         -0.10156    0.03123 3022.13238  -3.252  0.00116
##
## (Intercept)                      **
## surprise                        ***
## conditionHuman-Human              *
## conditionAI-AI                   ***
## surprise:conditionHuman-Human
## surprise:conditionAI-AI          **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) surprs cndH-H cAI-AI sr:H-H
## surprise      0.772
## cndtnHmn-Hm   -0.525 -0.405
## condtnAI-AI   -0.608 -0.469  0.319
## srprs:cnH-H   -0.367 -0.476  0.773  0.223
## srprs:AI-AI   -0.472 -0.611  0.247  0.821  0.291
##
## Condition-specific novelty-to-resonance slopes:
##   condition  surprise.trend      SE df asymp.LCL asymp.UCL
## Human-AI      0.986 0.0191 Inf      0.948      1.023
## Human-Human    0.963 0.0353 Inf      0.894      1.032
## AI-AI          0.884 0.0247 Inf      0.836      0.933
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95

```



11 Supplemental: Role-Level View



12 Summary

Primary cross-condition result: within each condition, $A_{\text{influence}} = |\beta_{\text{surprise, slot=1}} - \beta_{\text{surprise, slot=2}}|$ from $\text{resonance} \sim \text{surprise} * \text{slot} + (1|\text{conversation_id})$. The pooled LMM contributes only the unsigned magnitude column. Per-story OLS slope differences provide the label-invariant robustness test.

Robustness layers added:

- bootstrap 95% CIs on per-condition mean of per-story $A_{\text{influence}}$
- pairwise Cliff's δ effect sizes
- planned single-df contrast HA vs (HH+AA)/2

- length-controlled re-fit (+ `nwords`) of the unsigned magnitude

Supplemental views retained:

- condition-level marginal distributions and pooled `condition` LMMs
- pooled novelty-to-resonance model with `surprise * condition` interaction
- within-condition role box-plots (descriptive only; HA Human/AI inferential view lives in `analysis/human-ai/novelty.Rmd`)

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized asymmetry across all conditions)

Analysis completed: 2026-05-06 16:07:06.622933